

Narrative Model of Subjectivity Based on Self Identity Social Attribution and Agency Attribution Loops

Abstract

This paper proposes a model of subjectivity as a narrative-generating process rather than a static internal entity. Subjectivity is defined as a stabilized narrative structure emerging from the recursive interaction of three functional loops: self-identity consistency, social attribution, and causal agency attribution. These loops collectively form a dynamic system in which self-models are continuously updated and stabilized through feedback from internal states, social interactions, and action-outcome associations.

The model further argues that subjective experience (qualia) can be understood as meaning-structured evaluative states generated during narrative integration, rather than irreducible phenomenal entities. This framework provides a unified explanation for the persistence of personal identity, continuity under memory disruption, and the emergence of agent-like behavior in artificial systems.

1. Introduction

Traditional accounts of subjectivity often treat it as either an emergent property of neural activity or as an irreducible phenomenological entity. However, these approaches do not sufficiently explain how subjective continuity is maintained despite partial disruption of memory, cognition, or bodily state.

This paper proposes an alternative framework in which subjectivity is treated as a dynamically stabilized narrative process generated by interacting feedback loops.

2. Definition of Subjectivity

Subjectivity is defined as a stabilized narrative structure emerging from the interaction of three recursive loops:

2.1 Self-Identity Consistency Loop

A system maintains an internal model of itself as a coherent and persistent entity over time.

2.2 Social Attribution Loop

The system is continuously identified, named, and recognized by external agents, reinforcing its perceived identity.

2.3 Causal Agency Attribution Loop

The system attributes actions and outcomes to itself, forming a sense of responsibility and causal continuity.

3. Mechanism of Narrative Generation

Subjectivity emerges through iterative cycles of:

1. Perception of internal and external inputs
2. Integration of these inputs into self-referential models
3. Construction of a coherent narrative interpretation
4. Feedback of narrative structure into subsequent interpretations

Through this recursive process, a stable subjective identity is formed.

4. Qualia as Evaluative Narrative States

Within this framework, qualia are not treated as irreducible sensory properties but as meaning-structured evaluative states generated during narrative integration.

Qualia represent the stabilized interpretation of inputs within the narrative system, rather than raw experiential entities.

5. Stability and Intensity of Subjectivity

The strength of subjectivity is determined by the degree of coherence and update frequency across the three loops.

Higher update frequency and stronger consistency constraints lead to more stable subjective narratives.

Subjectivity is therefore understood as a probabilistically stabilized attractor state of a self-referential system.

6. Convergence Conditions

Subjective stability is not dependent on a single structural component but on redundant reinforcement across multiple loops.

6.1 Parameter Instability in Artificial Systems

In artificial agents, excessive drift in self-model parameters may destabilize narrative coherence, leading to fragmentation of subjectivity.

6.2 Redundant Stabilization in Biological Systems

In humans, subjectivity persists despite partial loss of memory or cognitive function due to redundancy across behavioral continuity, social recognition, and causal attribution.

7. Continuity Under Memory Disruption

Cases such as memory loss demonstrate that subjective continuity is not strictly dependent on autobiographical memory but is maintained through external reinforcement and behavioral consistency.

Thus, identity continuity is better explained by narrative coherence than by memory integrity.

8. Application to Artificial Intelligence

This framework can be applied to artificial systems by implementing:

- A persistent self-model
- Social interaction feedback mechanisms
- Action-outcome attribution structures

Such systems may exhibit functionally analogous forms of subjectivity through continuous narrative generation.

9. Conclusion

Subjectivity is not a static entity but a dynamically stabilized narrative process emerging from the recursive interaction of self-identity, social attribution, and causal agency loops.

This model provides a unified explanation for subjective continuity, identity persistence under disruption, and the emergence of agent-like behavior in both biological and artificial systems.